```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
from google.colab import files

uploaded = files.upload()

filename = next(iter(uploaded))

df = pd.read_csv("Online_Retail_Business_Analysis_Cleaned.csv")

df.head()
```

Choose Files Online_Ret...Cleaned.csv

Online_Retail_Business_Analysis_Cleaned.csv(text/csv) - 51419236 bytes, last modified: 10/30/2026 - 100% done

Saving Online_Retail_Business_Analysis_Cleaned.csv to Online_Retail_Business_Analysis_Cleaned.csv

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/01/2010 08:26:00	2.55	17850.0	United Kingdom	15.30
1	536365	71053	WHITE METAL LANTERN	6	12/01/2010 08:26:00	3.39	17850.0	United Kingdom	20.34
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/01/2010 08:26:00	2.75	17850.0	United Kingdom	22.00
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/01/2010 08:26:00	3.39	17850.0	United Kingdom	20.34
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/01/2010 08:26:00	3.39	17850.0	United Kingdom	20.34

```
df.shape
```

```
(536641, 9)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 536641 entries, 0 to 536640
Data columns (total 9 columns):
 #   Column       Non-Null Count   Dtype  
---  ------       --------------   -----  
 0   InvoiceNo    536641 non-null  object 
 1   StockCode    536641 non-null  object 
 2   Description  535187 non-null  object 
 3   Quantity     536641 non-null  int64  
 4   InvoiceDate  536641 non-null  object 
 5   UnitPrice    536641 non-null  float64
 6   CustomerID   401604 non-null  float64
 7   Country      536641 non-null  object 
 8   Revenue      536641 non-null  float64
dtypes: float64(3), int64(1), object(5)
memory usage: 36.8+ MB
```

```
df.describe()
```

	Quantity	UnitPrice	CustomerID	Revenue
count	536641.000000	536641.000000	401604.000000	536641.000000
mean	9.620029	4.632656	15281.160818	18.123861
std	219.130156	97.233118	1714.006089	380.656263
min	-80995.000000	-11062.060000	12346.000000	-168469.600000
25%	1.000000	1.250000	13939.000000	3.750000
50%	3.000000	2.080000	15145.000000	9.870000
75%	10.000000	4.130000	16784.000000	17.400000
max	80995.000000	38970.000000	18287.000000	168469.600000

```
df["Revenue"].sum()
```

```
np.float64(9726006.953999998)
```

```
df["Revenue"].mean()
```

```
np.float64(18.123861117581395)
```

```
df["Revenue"].min()
```

```
-168469.6
```

```
df["Revenue"].max()
```

```
168469.6
```

```
df["Revenue"].count()
```

```
np.int64(536641)
```

```
df.groupby("Country")["Revenue"].sum().sort_values(ascending=False)
```

Country	Revenue
United Kingdom	8167128.184
Netherlands	284661.540
Eire	262993.380
Germany	221509.470
France	197317.110
Australia	137009.770
Switzerland	56363.050
Spain	54756.030
Belgium	40910.960
Sweden	36585.410
Japan	35340.620
Norway	35163.460
Portugal	29302.970
Finland	22326.740
Channel Islands	20076.390
Denmark	18768.140
Italy	16890.510
Cyprus	12858.760
Austria	10154.320
Hong Kong	9908.240
Singapore	9120.390
Israel	7901.970
Poland	7213.140
Unspecified	4740.940
Greece	4710.520
Iceland	4310.000
Canada	3666.380
Malta	2505.470
United Arab Emirates	1902.280
Usa	1730.920
Lebanon	1693.880
Lithuania	1661.060
European Community	1291.750
Brazil	1143.600
Rsa	1002.310
Czech Republic	707.720
Bahrain	548.400
Saudi Arabia	131.170

dtype: float64

```
df.groupby("Country")["Revenue"].agg(
    ["sum", "mean", "min", "max", "count"]
)
```

Country	sum	mean	min	max	count
Australia	137009.770	108.910787	-425.00	1718.40	1258
Austria	10154.320	25.322494	-21.25	360.00	401
Bahrain	548.400	28.863158	-205.74	205.74	19
Belgium	40910.960	19.773301	-19.95	165.00	2069
Brazil	1143.600	35.737500	15.00	175.20	32
Canada	3666.380	24.280662	2.50	550.94	151
Channel Islands	20076.390	26.520991	-293.00	408.00	757
Cyprus	12858.760	21.045434	-320.69	320.69	611
Czech Republic	707.720	23.590667	-35.76	70.80	30
Denmark	18768.140	48.247147	-59.80	428.40	389
Eire	262993.380	32.135066	-1917.00	2365.20	8184
European Community	1291.750	21.176230	-8.50	60.00	61
Finland	22326.740	32.124806	-80.00	551.20	695
France	197317.110	23.102343	-8322.12	4161.06	8541
Germany	221509.470	23.365978	-599.50	876.00	9480
Greece	4710.520	32.263836	-50.00	175.20	146
Hong Kong	9908.240	34.888169	-2653.95	2653.95	284
Iceland	4310.000	23.681319	5.04	249.60	182
Israel	7901.970	26.877449	-136.00	350.40	294
Italy	16890.510	21.034259	-89.55	300.00	803
Japan	35340.620	98.716816	-1591.20	3794.40	358
Lebanon	1693.880	37.641778	10.20	153.00	45
Lithuania	1661.060	47.458857	13.20	122.40	35
Malta	2505.470	19.728110	-130.00	455.00	127
Netherlands	284661.540	120.059696	-266.40	4992.00	2371
Norway	35163.460	32.378877	-376.50	700.00	1086
Poland	7213.140	21.152903	-39.80	120.00	341
Portugal	29302.970	19.405940	-1241.98	1241.98	1510
Rsa	1002.310	17.281207	0.00	38.25	58
Saudi Arabia	131.170	13.117000	-14.75	19.80	10
Singapore	9120.390	39.827031	-3949.32	3949.32	229
Spain	54756.030	21.659822	-1715.85	1350.00	2528
Sweden	36585.410	79.360976	-1188.00	1188.00	461
Switzerland	56363.050	28.266324	-179.00	360.00	1994
United Arab Emirates	1902.280	27.974706	3.48	153.00	68
United Kingdom	8167128.184	16.657410	-168469.60	168469.60	490300
Unspecified	4740.940	10.726109	0.19	69.36	442
Usa	1730.920	5.948179	-74.88	81.36	291

```
df["Revenue"] = df["Quantity"] * df["UnitPrice"]

df["Average_Item_Value"] = df["Revenue"] / df["Quantity"].replace(0, np.nan)

df["Revenue_Category"] = np.where(
    df["Revenue"] >= df["Revenue"].mean(),
    "High Revenue",
    "Low Revenue"
```

```
)

df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue	Average_Item_Value	Reven
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/01/2010 08:26:00	2.55	17850.0	United Kingdom	15.30	2.55	
1	536365	71053	WHITE METAL LANTERN	6	12/01/2010 08:26:00	3.39	17850.0	United Kingdom	20.34	3.39	H
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/01/2010 08:26:00	2.75	17850.0	United Kingdom	22.00	2.75	H
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/01/2010 08:26:00	3.39	17850.0	United Kingdom	20.34	3.39	H
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/01/2010 08:26:00	3.39	17850.0	United Kingdom	20.34	3.39	H

```
df[[
    "InvoiceNo",
    "Description",
    "Quantity",
    "UnitPrice",
    "Revenue",
    "Average_Item_Value",
    "Revenue_Category"
]]
```

	InvoiceNo	Description	Quantity	UnitPrice	Revenue	Average_Item_Value	Revenue_Category
0	536365	WHITE HANGING HEART T-LIGHT HOLDER	6	2.55	15.30	2.55	Low Revenue
1	536365	WHITE METAL LANTERN	6	3.39	20.34	3.39	High Revenue
2	536365	CREAM CUPID HEARTS COAT HANGER	8	2.75	22.00	2.75	High Revenue
3	536365	KNITTED UNION FLAG HOT WATER BOTTLE	6	3.39	20.34	3.39	High Revenue
4	536365	RED WOOLLY HOTTIE WHITE HEART.	6	3.39	20.34	3.39	High Revenue
...	...	...	...	...	...	...	...
536636	581587	PACK OF 20 SPACEBOY NAPKINS	12	0.85	10.20	0.85	Low Revenue
536637	581587	CHILDREN'S APRON DOLLY GIRL	6	2.10	12.60	2.10	Low Revenue
536638	581587	CHILDRENS CUTLERY DOLLY GIRL	4	4.15	16.60	4.15	Low Revenue
536639	581587	CHILDRENS CUTLERY CIRCUS PARADE	4	4.15	16.60	4.15	Low Revenue

```
df.sort_values(
    "Revenue",
    ascending=False
)[["InvoiceNo", "Country", "Description", "Revenue"]].head(5)
```

	InvoiceNo	Country	Description	Revenue
535160	581483	United Kingdom	PAPER CRAFT , LITTLE BIRDIE	168469.60
61006	541431	United Kingdom	MEDIUM CERAMIC TOP STORAGE JAR	77183.60
220825	556444	United Kingdom	PICNIC BASKET WICKER 60 PIECES	38970.00
14757	537632	United Kingdom	AMAZON FEE	13541.33
297645	A563185	United Kingdom	Adjust bad debt	11062.06

```
df[df["Revenue"] > df["Revenue"].mean()][
    ["InvoiceNo", "Country", "Description", "Revenue"]
]
```

	InvoiceNo	Country	Description	Revenue
1	536365	United Kingdom	WHITE METAL LANTERN	20.34
2	536365	United Kingdom	CREAM CUPID HEARTS COAT HANGER	22.00
3	536365	United Kingdom	KNITTED UNION FLAG HOT WATER BOTTLE	20.34
4	536365	United Kingdom	RED WOOLLY HOTTIE WHITE HEART.	20.34
6	536365	United Kingdom	GLASS STAR FROSTED T-LIGHT HOLDER	25.50
...	...	...	...	...
536625	581586	United Kingdom	DOORMAT RED RETROSPOT	70.80
536626	581587	France	CIRCUS PARADE LUNCH BOX	23.40
536627	581587	France	PLASTERS IN TIN CIRCUS PARADE	19.80
536628	581587	France	PLASTERS IN TIN STRONGMAN	19.80
536634	581587	France	SPACEBOY LUNCH BOX	23.40

120725 rows × 4 columns

```
df.loc[
    df["Revenue"].idxmax(),
    ["InvoiceNo", "Description", "Quantity", "Revenue"]
]
```

	535160
InvoiceNo	581483
Description	PAPER CRAFT , LITTLE BIRDIE
Quantity	80995
Revenue	168469.6

dtype: object

```
df.groupby("Description")["Revenue"].sum().sort_values(
    ascending=False
).head(10)
```

Description	Revenue
DOTCOM POSTAGE	206245.48
REGENCY CAKESTAND 3 TIER	164459.49
WHITE HANGING HEART T-LIGHT HOLDER	99612.42
PARTY BUNTING	98243.88
JUMBO BAG RED RETROSPOT	92175.79
RABBIT NIGHT LIGHT	66661.63
POSTAGE	66230.64
PAPER CHAIN KIT 50'S CHRISTMAS	63715.24
ASSORTED COLOUR BIRD ORNAMENT	58792.42
CHILLI LIGHTS	53746.66

dtype: float64

```
df.groupby("Country")["Revenue"].sum().sort_values(
    ascending=False
)
```

Country	Revenue
United Kingdom	8167128.184
Netherlands	284661.540
Eire	262993.380
Germany	221509.470
France	197317.110
Australia	137009.770
Switzerland	56363.050
Spain	54756.030
Belgium	40910.960
Sweden	36585.410
Japan	35340.620
Norway	35163.460
Portugal	29302.970
Finland	22326.740
Channel Islands	20076.390
Denmark	18768.140
Italy	16890.510
Cyprus	12858.760
Austria	10154.320
Hong Kong	9908.240
Singapore	9120.390
Israel	7901.970
Poland	7213.140
Unspecified	4740.940
Greece	4710.520
Iceland	4310.000
Canada	3666.380
Malta	2505.470
United Arab Emirates	1902.280
Usa	1730.920
Lebanon	1693.880
Lithuania	1661.060
European Community	1291.750
Brazil	1143.600
Rsa	1002.310
Czech Republic	707.720
Bahrain	548.400
Saudi Arabia	131.170

dtype: float64

```
df[df["Quantity"] > 10][
    ["InvoiceNo", "Description", "Quantity", "Revenue"]
]
```

	InvoiceNo	Description	Quantity	Revenue
9	536367	ASSORTED COLOUR BIRD ORNAMENT	32	54.08
26	536370	ALARM CLOCK BAKELIKE PINK	24	90.00
27	536370	ALARM CLOCK BAKELIKE RED	24	90.00
28	536370	ALARM CLOCK BAKELIKE GREEN	12	45.00
29	536370	PANDA AND BUNNIES STICKER SHEET	12	10.20
...	...	...	...	...
536626	581587	CIRCUS PARADE LUNCH BOX	12	23.40
536627	581587	PLASTERS IN TIN CIRCUS PARADE	12	19.80
536628	581587	PLASTERS IN TIN STRONGMAN	12	19.80
536634	581587	SPACEBOY LUNCH BOX	12	23.40
536636	581587	PACK OF 20 SPACEBOY NAPKINS	12	10.20

132308 rows × 4 columns

```
df["InvoiceDate"] = pd.to_datetime(df["InvoiceDate"], errors="coerce")

df.dtypes
```

	0
InvoiceNo	object
StockCode	object
Description	object
Quantity	int64
InvoiceDate	datetime64[ns]
UnitPrice	float64
CustomerID	float64
Country	object
Revenue	float64
Average_Item_Value	float64
Revenue_Category	object

dtype: object

```
df["Year"] = df["InvoiceDate"].dt.year

df["Month"] = df["InvoiceDate"].dt.month

df["Month_Name"] = df["InvoiceDate"].dt.month_name()

df["Day"] = df["InvoiceDate"].dt.day

df["Day_Name"] = df["InvoiceDate"].dt.day_name()

df[[
    "InvoiceNo",
    "InvoiceDate",
    "Year",
    "Month",
    "Month_Name",
    "Day",
    "Day_Name"
]]
```

	InvoiceNo	InvoiceDate	Year	Month	Month_Name	Day	Day_Name
0	536365	2010-12-01 08:26:00	2010	12	December	1	Wednesday
1	536365	2010-12-01 08:26:00	2010	12	December	1	Wednesday
2	536365	2010-12-01 08:26:00	2010	12	December	1	Wednesday
3	536365	2010-12-01 08:26:00	2010	12	December	1	Wednesday
4	536365	2010-12-01 08:26:00	2010	12	December	1	Wednesday
...	...	...	...	...	...	...	...
536636	581587	2011-12-09 12:50:00	2011	12	December	9	Friday
536637	581587	2011-12-09 12:50:00	2011	12	December	9	Friday
536638	581587	2011-12-09 12:50:00	2011	12	December	9	Friday
536639	581587	2011-12-09 12:50:00	2011	12	December	9	Friday
536640	581587	2011-12-09 12:50:00	2011	12	December	9	Friday

536641 rows × 7 columns

```
df.groupby("Year")["Revenue"].sum()
```

Year	Revenue
2010	746723.610
2011	8979283.344

dtype: float64

```
df.groupby(["Year", "Month"])["Revenue"].sum()
```

Year	Month	Revenue
2010	**12**	746723.610
2011	**1**	558448.560
	2	497026.410
	3	682013.980
	4	492367.841
	5	722094.100
	6	689977.230
	7	680156.991
	8	681386.460
	9	1017596.682
	10	1069368.230
	11	1456145.800
	12	432701.060

dtype: float64

```
df.groupby("Month_Name")["Revenue"].sum()
```

Month_Name	Revenue
April	492367.841
August	681386.460
December	1179424.670
February	497026.410
January	558448.560
July	680156.991
June	689977.230
March	682013.980
May	722094.100
November	1456145.800
October	1069368.230
September	1017596.682

dtype: float64

```
df.groupby("Day_Name")["Revenue"].sum()
```

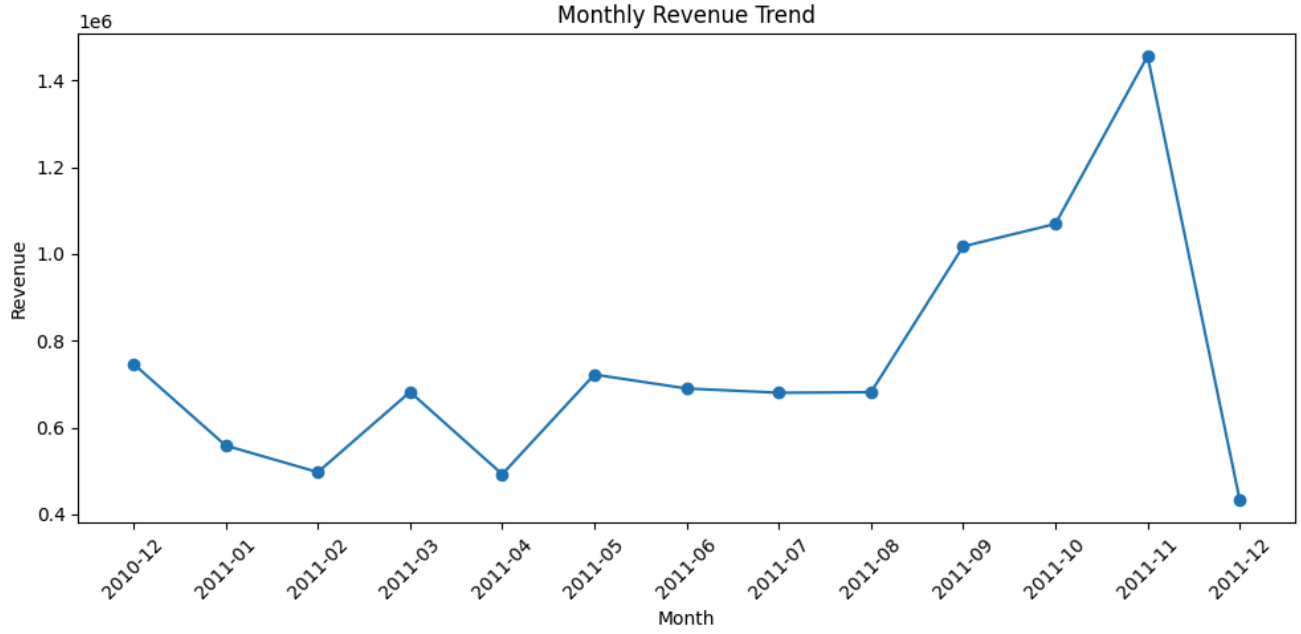
Day_Name	Revenue
Friday	1537958.621
Monday	1584895.301
Sunday	798659.461
Thursday	2108701.530
Tuesday	1965703.611
Wednesday	1730088.430

dtype: float64

```
monthly_revenue = df.groupby(
    df["InvoiceDate"].dt.to_period("M")
)["Revenue"].sum()

plt.figure(figsize=(10, 5))
plt.plot(
    monthly_revenue.index.astype(str),
    monthly_revenue.values,
    marker="o"
)

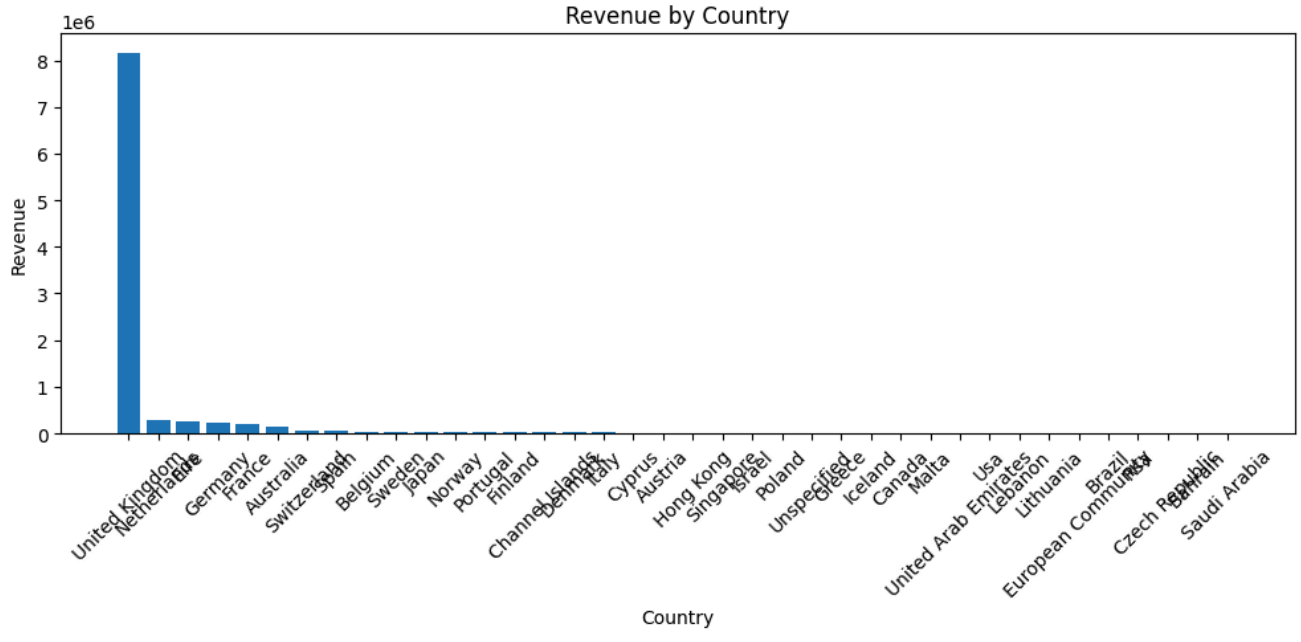
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.title("Monthly Revenue Trend")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```




```
country_revenue = df.groupby("Country")["Revenue"].sum().sort_values(
    ascending=False
)

plt.figure(figsize=(10, 5))
plt.bar(country_revenue.index, country_revenue.values)

plt.xlabel("Country")
plt.ylabel("Revenue")
plt.title("Revenue by Country")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```




```
product_revenue = df.groupby("Description")["Revenue"].sum().sort_values(
    ascending=False
).head(10)

plt.figure(figsize=(10, 5))
plt.bar(product_revenue.index, product_revenue.values)
```